Scleroderma is one of the collagen vascular diseases that is characterized by a pattern of inflammation, vascular sclerosis, and fibrosis of the skin and organs. The etiology is not known, but the pathophysiology of the disease progresses as follows:

1. *Injury to the vascular endothelium*
2. *Vascular obliteration and leakage of proteins into the interstitial space*
3. *Tissue edema and lymphatic obstruction due to the protein leakage*
4. *Tissue fibrosis*

As indicated in the table below, scleroderma affects virtually every system in the body and presents a host of potential anesthetic difficulties. Positioning may be difficult due to flexion contractures (especially of the hands). The patient may require a fiberoptic intubation for a difficult airway due to decreased mandibular motion. Pharyngeal telangiectasis may increase the likelihood of bleeding due to airway manipulation. Chronic hypertension result in a contracted intravascular volume. Hypotonia of the lower esophageal sphinter places them at risk for gastric reflux. They are often prone to dry eyes, making corneal abrasion more likely to occur. It is also important to avoid factors such as acidosis and arterial hypoxemia that may worsen pulmonary hypertension. When possible, regional anesthesia offers the advantage of peripheral vasodilation and improved postoperative analgesia.

System	Complications
Integument	Nonpitting edema, taut skin leading to contractures
Muscular	Myopathy with proximal skeletal muscle weakness. Elevated creatine kinase
Skeletal	Avascular necrosis of the femoral head, arthritis
Nervous System	Peripheral and cranial nerve neuropathy from nerve compression
Cardiovascular	Systemic and pulmonary hypertension due to vascular sclerosis, cardiac conduction abnormalities, fibrosis of the myocardium, Raynaud's phenomenon (in 95% of cases)
Pulmonary	Decreased diffusion capacity, arterial hypoxemia, decreased pulmonary compliance due to fibrosis (chest wall compliance is not affected significantly)
Renal	Renal artery stenosis, malignant hypertension, renal failure (note:corticosteroids can precipitate a renal crisis in patients with scleroderma)
Gastrointestinal	Hypomotility of the GI tract, decreased lower esophageal sphincter pressure and increased risk of reflux

Hines RL, Marschall KE. Stoeltings Anesthesia & Co-Existing Diseases. 6th ed. New York, NY: Churchill-Livingston; 2012: 441-443.

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